

Mixed-signal modelling and verification of the CAN bus in eSim

Sourabh Kumar

Protocol and motivation

I propose to model a CAN (Controller Area Network) bus in eSim, with the physical layer in Ngspice and the protocol controller in VHDL through NGHDL.

I chose CAN over UART or SPI because it needs both halves of eSim to work together. Arbitration in CAN is not a logical operation on a data structure: a node drives a recessive level, reads the actual voltage back off the wire, and if it reads dominant it has lost and must stop. That only means something if the analog bus is really there, with real thresholds and real driver impedance. The same is true of the acknowledge slot and of error signalling, where a node deliberately violates the bit-stuffing rule to destroy a frame everyone else is receiving.

Physical layer

A differential pair with 45 ohm driver on-resistance, 120 ohm termination at both ends, 500 pF per line, and a 10 kohm bias network holding the recessive state at 2.5 V on both rails. The receiver is a differential comparator centred at 0.7 V, between the 0.5 V recessive and 0.9 V dominant thresholds of ISO 11898-2. A transmission-line element represents 220 m of twisted pair at 120 ohm and 2e8 m/s.

Controller

Bit timing with hard synchronisation and resynchronisation, bit stuffing and destuffing, CRC-15 with polynomial 0x4599, frame assembly and decoding, arbitration, acknowledgement, error frames, and the transmit and receive error counters that move a node between error-active, error-passive and bus-off.

Bit timing

125 kbit/s, so an 8.000 us bit divided into 16 time quanta of 500 ns, giving a 2.000 MHz controller clock. The segments are 1 / 5 / 6 / 4 tq with the sample point at 12 tq, or 75 per cent.

The propagation segment is not a guess. Over the modelled 220 m bus the one-way delay measures 1.1136 us and the transceiver adds 21.6 ns, so the round trip is 2.27 us against the 2.500 us that five time quanta allow - a margin of 9 per cent. Placing the sample point at 75 per cent puts it more than one round trip after the edge, which is also why reflections from a badly terminated bus have decayed before the bit is read.

What the simulation will show

Four nodes with identifiers 0x0A5, 0x123, 0x2AA and 0x555 request the bus at the same instant. The identifiers are chosen so the outcome is predictable from their bit patterns: 0x555 differs first, at identifier bit 10, then 0x2AA at bit 9, then 0x123 at bit 8. If arbitration works, the three losers withdraw exactly one bit time apart and the node with 0x0A5 transmits its frame to completion while the others receive it.

For error handling I inject a physical fault - a low-resistance short across the differential pair - and show that the affected node counts bit errors, signals them, and eventually removes itself from the bus, while a healthy node on the same wire is unaffected.

Verification

Each block is checked against a reference model written from the ISO specification rather than from my own code, so the two can disagree. The CRC is compared over several hundred random frames, and frame assembly is compared bit for bit against independently generated expected output including the positions of stuff bits.

References

ISO 11898-1:2015, Road vehicles - Controller area network (CAN) - Part 1: Data link layer and physical signalling.

ISO 11898-2:2016, Part 2: High-speed medium access unit.
Bosch CAN Specification Version 2.0, Robert Bosch GmbH, 1991.